



The effect of driving demands on distraction engagement and glance behaviors: Results from naturalistic data

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ABSTRACT

Objective: To understand drivers' engagement in distractions and their visual attention allocation under different driving demands. **Background:** Although distraction increases crash risk, drivers engage in distractions frequently with no negative consequences, likely in part due to their self-regulating behaviors. Prior research revealed a variety of self-regulating behaviors specifically related to cell-phone engagement, but very limited research has investigated whether and how driving demands affect engagement in distractions in general, particularly within a natural setting rather than in the simulator. **Method:** We used the Naturalistic Engagement in Secondary Tasks (NEST) dataset, a subset of SHRP2 data, to analyze secondary task engagement and off-path (not in direction of travel) eye glances. In addition to assessing their relation to environmental demand, we also considered driver age and chosen speed. **Results:** Higher visual difficulty environments (characterized as visually complex and/or with low visibility) were associated with a decreased likelihood of secondary task engagement as well as a decrease in off-path glances, particularly longer ones (>2s). Drivers 35 and older had lower rates of off-path glances compared to younger drivers. An increase in speed was associated with a decrease in the likelihood of task engagement in higher motor control difficulty environments (characterized as poor surface condition and/or curved road) but not in lower ones. **Conclusion:** Drivers appear to modulate their task engagement and off-path glances based on driving demands. However, given that inopportune short off-path glances can lead to crashes, interventions are still needed to help drivers better modulate their distraction engagement.

1. Introduction

Distracted driving is a leading concern for traffic safety. The latest report on U.S. annual crash data attributes 3166 fatalities (9% of total road fatalities) to distraction-affected crashes (National Highway Traffic Safety Administration, 2019). Distraction has also been shown to degrade driving performance in simulator and on-road studies, with increased brake response times to hazards (e.g., D'Addario and Donmez, 2019), decreased time to collision (e.g., Chen et al., 2018), and larger variability in lane keeping (e.g., Engström et al., 2005). The largest naturalistic driving study to date, the Second Strategic Highway Research Program 2 (SHRP2), found that 68% of crashes recorded involved some form of observable distraction (Dingus et al., 2016). Moreover, the SHRP2 study estimated that drivers were distracted approximately 50% of the time (Dingus et al., 2016).

Distraction, however, does not always lead to crashes. Drivers often

divide their attention between secondary tasks and the driving task with little to no detriment to their performance and safety (Dingus et al., 2016). Observations from naturalistic studies suggest that drivers may self-regulate their distraction engagement depending on the driving context. For example, drivers are more likely to initiate cell-phone conversations or engage in visual-manual phone tasks when standing still in traffic, compared to when driving at high speeds (Christoph et al., 2019; Funkhouser and Sayer, 2012; Tivesten and Dozza, 2015). Klauer et al. (2006) reported descriptive statistics from the 100-Car Study data on driver engagement in secondary tasks in different environmental conditions. Increased secondary task engagement was observed in less demanding environments, such as driving in daylight or in clear weather, divided highways, straight roads, and dry roads.

Another consideration regarding self-regulation of driver distraction is individual differences. Distraction engagement and the associated crash risk may vary with driver characteristics like age and experience.

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In the U.S., drivers 20 years old and younger disproportionately account for a higher percentage of drivers involved in distraction-affected crashes (Singh, 2010). A survey study focused on technology-based distractions (e.g., cell phones and in-vehicle systems) found that perceived social norms is a significant factor for drivers 30 and younger, whereas personality, not social norms, is a significant factor for drivers above 30 (Chen and Donmez, 2016). In comparison to novice drivers, experienced (typically older) drivers exhibited more proactive responses (anticipatory behaviors) to changes in traffic in a driving simulator study and were more efficient in how they scanned the driving environment and how they divided their attention between a visual-manual distraction and the roadway (He and Donmez, 2018).

In non-naturalistic studies that are conducted as controlled experiments, driver distraction is generally studied by requiring participants to engage in specific secondary tasks. While such an experimental paradigm cannot assess self-regulation of distraction engagement, it does generate insights on risk-reducing driving behaviors when drivers are engaged in secondary tasks. For example, simulator studies have found that when participants were distracted by a cell-phone conversation, they were less likely to overtake slower vehicles (Charlton, 2009) or change lanes (Cooper et al., 2009), and more likely to increase their following distance to a lead vehicle (e.g., Strayer and Drews, 2004) or reduce speed more substantially when negotiating curves (Oviedo-Trespalacios et al., 2017). On the other hand, when participants in a simulator study were allowed to freely pace their engagement with a tablet-based secondary task, large variability was found in how often participants chose to engage—from none at all to continual engagement—in the contrived task, and male participants showed a higher level of distraction engagement compared to their female counterparts (Chen et al., 2018).

Naturalistic driving data, which relies on sensors installed in participants' own vehicles to capture everyday driving behaviors, is particularly useful for investigating the wide range of conditions in which drivers may engage, to various extents, in secondary tasks. In particular, drivers' glance behaviors can help provide a more accurate picture of how drivers allocate their attention to the roadway and the associated risks. Multiple naturalistic studies have demonstrated that off-path glances, i.e., glances away from the forward roadway, are associated with an increased crash risk; longer glances (>2 s) are particularly detrimental to safety (Klauer et al., 2006; Simons-Morton et al., 2014), but even short glances can lead to crashes (Victor et al., 2015). In line with this finding, visual-manual secondary tasks have been consistently found to be the most detrimental to safety compared to other types of secondary tasks, given that driving itself is a highly visual-manual task (Dingus et al., 2016; Klauer et al., 2006).

Literature has shown that drivers modify their visual attention allocation based on their expectations (Räsänen and Summala, 1998; Tijerina et al., 2004) and driving demands (Shinar, 2008; Tivesten and Dozza, 2014). For example, Tivesten and Dozza (2014) found that drivers in a naturalistic driving study spent more time looking at the road and exhibited fewer long off-road glances when turning and when a lead or oncoming vehicle is present. These changes in glance patterns occurred whether or not drivers were engaged in a visual-manual phone

task. Further, when drivers were engaged in a phone task, longer off-road glances were associated with lower speeds.

These naturalistic observations on self-regulatory distraction engagement behaviors have largely focused on cell-phone engagement. The gap remains in understanding to what extent drivers adapt their general distraction engagement—including but not limited to mobile device use—to demands imposed by the driving environment. In this paper, we utilized the Naturalistic Engagement in Secondary Tasks (NEST) dataset (Owens et al., 2015), part of the SHRP2 naturalistic driving data collection, to investigate the role of driving demands in the self-regulation of driver distraction and in visual attention allocation (i.e., off-path glances, or glances away from the direction of travel). Secondary task engagement and off-path glances were analyzed with environmental demand, GPS speed, and driver age as explanatory variables. We classified the environment as having higher/lower visual difficulty (based on visual complexity and visibility) and higher/lower motor control difficulty (based on road curvature and surface condition). Speed was included to better capture driving demands, given that environmental difficulty (or demand) is not the sole component of driving demand. Further, driver age was adopted to capture the potential variability in self-regulatory behaviors and driving skills that different age groups may exhibit.

2. Methods

2.1. NEST dataset

Our analysis utilized baseline data (i.e., data with no near-crash or crash events) from the NEST dataset (Owens et al., 2015), which was derived from SHRP2 data by the Virginia Technical Transportation Institute (VTTI). The SHRP2 study monitored approximately 3400 drivers during a 3-year period between 2010 and 2013 (Dingus et al., 2015). About 50 million miles of naturalistic driving data were collected from a nationally representative sample across six states in the United States: Washington, Florida, New York, North Carolina, Pennsylvania, and Indiana. NEST was sampled from the data of SHRP2 drivers who had at least one safety critical event (i.e., near-crash or crash event) for which secondary task engagement was identified as a potential contributing factor. Thus, NEST consists of a sample representing drivers likely at an elevated risk for distraction-affected crashes. Overall, NEST has 944 baselines associated with 204 drivers, with 4 to 16 baselines per driver. NEST data was coded in more detail beyond the standard SHRP2 data to provide a richer description for investigating driver distraction. Specifically, each NEST baseline was coded over a 20 s period as opposed to the 6 s period used for SHRP2 data reduction. Some variables were coded once per baseline, such as weather, whereas some variables were coded twice per baseline (for each 10 s period), such as traffic density. Table 1 provides further details about the coding of NEST variables that were used in our analysis. No drug or alcohol impairment was noted for NEST baselines.

2.2. Model variables

As stated earlier, our objective was to investigate how and to what extent drivers adapt their secondary task engagement and allocate visual

Table 1
NEST variables used in our analysis and their coding rates. Each baseline was approximately 20 s.

Variable Type	NEST Variables	Coding Rate
Environmental	Weather (e.g., no adverse conditions, fog, snowing)	Once per baseline
	Lighting (e.g., darkness, dusk, daylight)	Twice per baseline
	Locality (e.g., residential, urban)	
	Traffic density (e.g., free flow, flow with some restrictions)	
	Surface condition (e.g., wet, dry, snowy)	
	Alignment (e.g., straight, curve left)	
Behavioral	Secondary task (e.g., drinking, dancing, texting on cell phone, see Table 2 for a full list)	Up to six secondary tasks were coded twice per baseline
	Vehicle speed (km/h)	At 10 Hz (~200 frames per baseline)
	Drivers' glance location (e.g., forward, right windshield, cell phone, see Table 3 for a full list)	

attention based on driving demands. The following sections describe our dependent and explanatory variables.

2.2.1. Dependent variables

Secondary Task Engagement. “Secondary task engagement” was a dichotomous variable indicating whether or not a driver was engaged in a secondary task during a given baseline. There were 40 different secondary tasks observed in NEST baselines (Table 2). Although NEST listed “the presence of passengers/children with no interaction” as secondary tasks, we did not consider them as secondary task engagement. Moreover, secondary task types that were coded as unknown or N/A were excluded from our analysis.

Glance Metrics. Table 3 presents the NEST glance location categories. Our glance analyses were performed in a manner similar to other researchers who have used data from SHRP2 NDS (Victor et al., 2015), as well as its predecessors, the 100-Car (Klauer et al., 2006) and the

Table 2
NEST secondary task categories, and the number of baselines a given secondary task was observed in.

Secondary Task Type	NEST Secondary Tasks	No. of Baselines
Dancing/singing	Talking/singing	174
	Dancing	33
Drinking/eating	Drinking	21
	Eating	18
Interaction with carried-in device	Cell phone, other	91
	Texting on cell phone	85
	Locating/reaching/answering cell phone	43
	Operating PDA/other handheld device	3
	Viewing PDA/other handheld device	3
	Dialing hand-held cell phone	2
	Looking at an object external to the vehicle	87
Outside distraction	Other external distraction	77
	Looking at pedestrian	5
	Passenger in adjacent seat – interaction	171
	Passenger in rear seat – interaction	22
	Child in rear seat – interaction	7
Passenger interaction	Child in adjacent seat – interaction	5
	Other personal hygiene	26
	Biting nails/cuticles	25
	Combing/brushing/fixing hair	8
	Removing/inserting/adjusting contact lenses or glasses	7
Personal hygiene	Removing/adjusting jewelry	4
	Applying make-up	2
	Brushing/flossing teeth	1
	Pet in vehicle	6
	Reaching/manipulating object	41
Pet in vehicle	Object in vehicle, other	20
	Reaching for object, other	18
	Reaching for/Lighting/Smoking/Extinguishing cigarette	10
	Reaching for food-related or drink-related item	2
	Moving object in vehicle	2
Reaching/manipulating object	Reaching for object that is a manufacturer-installed device	2
	Reaching for personal body-related item	2
	Talking/listening on cell phone	47
	Adjusting/monitoring radio	75
	Adjusting/monitoring climate control	20
Talking on hand-held cell phone	Adjusting/monitoring other devices integral to vehicle	15
	Inserting/retrieving CD	1
Visual/manual interaction with vehicle/integrated device/control		

Naturalistic Teenage Driving studies (Simons-Morton et al., 2014). That is, we focused on whether a glance was off-path (away from the forward roadway in direction of travel) or on-path. We were also interested in whether a glance was driving-related or not. For example, Tivesten and Dozza (2014) analyzed glances as being off-road vs. on-road, where on-road glances included glances to the forward roadway, as well as glances to intersecting roads and other road users that were relevant to the traffic situation. However, we were informed by VTTI that the glance coding performed on the SHRP2 NDS, and hence the NEST data, was directional rather than contextual. Thus, there are some glances which may or may not be considered as driving related (e.g., an over-the-shoulder check can be performed as a blind-spot check or to look at a billboard). Therefore, we grouped glance locations into “non-driving-related” and “potentially driving-related” categories as in Table 3. We conducted analyses on both the on/off-path categorization and the non/potentially-driving-related categorization. However, the latter is likely less reliable than the former given coding limitations.

Off-path glances were analyzed through four dependent variables: (1) percentage of time eyes were off-path, (2) number of off-path glances, (3) number of off-path glances > 1.6 s, and (4) number of off-path glances > 2 s. The same dependent variables were also analyzed for non-driving-related glances, in place of off-path glances. The glance duration thresholds (1.6 s and 2 s) adopted for our glance metrics have been used in other related research (e.g., Reimer et al., 2014). The 2 s threshold has been adopted by the National Highway Traffic Safety Administration as a safety-critical threshold for evaluating the visual-manual distraction potential of in-vehicle (2013) and portable (2016) devices, and is based on the finding that off-path glances longer than 2 s double the risk of crashes (Klauer et al., 2006). The 1.6 s threshold originates from in-vehicle task performance studies, which found that drivers do not, in general, direct their glances away from the forward roadway for longer than 1.6 s at a time (Wierwille, 1993). Glance durations were calculated following ISO 15007 (International Organization for Standardization, 2013), and we excluded the frames where the NEST glance location variable was coded as “no video”, “eyes closed”, “no eyes visible – glance location unknown”, and “other”.

2.2.2. Explanatory variables

The metrics presented above were analyzed with environmental demand, GPS speed, and driver age as explanatory variables. We classified the environment as having higher/lower visual difficulty (based on visual complexity and visibility) and higher/lower motor control difficulty (based on road curvature and surface condition). Speed was included to better capture driving demands, and driver age was adopted to capture the potentially different self-regulatory behaviors that different age groups may exhibit.

Visual and Motor Control Difficulty. Although NEST provides an extensive list of explanatory variables that can characterize environmental demands, given its relatively small sample size, it was infeasible to include all of these variables separately in our models. Hence, aggregations were performed to create meaningful groupings with reasonable sample sizes. It should be noted that we could not identify any earlier research that conducted similar aggregations, thus, the aggregations performed in this paper, as well as our other relevant publications (Kanaan et al., 2019; Risteska et al., 2018), were based on the subjective judgments of the authors.

In particular, environmental demand was captured through two explanatory variables: visual difficulty and motor control difficulty, each of which had two levels (lower and higher difficulty). The aggregations and the categorizations of the difficulty levels (lower vs. higher) were largely based on the NEST and SHRP2 variable descriptions; however, two of the authors also reviewed a subsample of the SHRP2 Insight Data Access Website video sequences of the forward road to inform the categorizations. Although this process is inherently subjective, (1) the categories are in relative terms (i.e., lower vs. higher) rather than absolute, and (2) when creating the categories, the authors were

Table 3

NEST glance location variables and their definitions.

Glance Direction	NEST Glance Location	Definition
Eyes-on-path	Forward (center)	Glance out the forward windshield directed towards the direction of the vehicle's travel.
	Left windshield	Glance out of the forward windshield where the driver appears to be looking specifically out of the left/right margin of the windshield (e.g., as if scanning for traffic before turning or glancing at oncoming traffic). The glance location is clearly not in the direction of travel (e.g., at road signs or buildings).
Eyes-off-path	Potentially driving related	Right windshield
	Non-driving related	Rearview mirror Left window/mirror Right window/mirror Over-the-shoulder Instrument cluster Center stack (control-bearing surfaces in the center of the front of the vehicle interior) Cell phone iPod (or similar) Interior object Passenger No eyes visible. Eyes are off-road.
		Glance toward the rearview mirror or equipment located around it. Glance to the left/right side mirror or window. Glance over either of the participant's shoulders. Glance to instrumented cluster underneath the dashboard. Glance to the vehicle's center stack (vertical). Glance at a cell phone, other electronic communication device, or cell-phone related equipment (e.g., chargers), no matter where it is located. Glance at an iPod or other personal digital music device no matter where it is located. Glance to an identifiable object in the vehicle other than a cell phone, including personal items brought in by participants, any part of their body, electronic devices other than cell phones, and OEM installed devices that don't fall into other categories. Glances to the center console are also included here. Glance to a passenger. Driver's eyes/face not visible. However, it is clear that the participant is not looking at the roadway.

blind to the dependent variables (i.e., secondary task engagement and glance metrics). A comprehensive description of the categorizations is presented in Risteska (2018).

Visual difficulty was based on the NEST variables of weather, lighting, locality, and traffic density. Overall, higher visual difficulty environments consisted of situations where visibility was low or where the environment was visually complex (e.g., high traffic density, complex locality with many traffic control devices and mixed modes of transport). Both visibility and visual complexity were seen as contributing to visual difficulty, given that fatality risk is four times higher at night (Williams, 2003) and that higher visual complexity of the environment interferes with visual search performance (Donderi, 2006).

Motor control difficulty was based on the NEST variables of surface condition and alignment, as crash risk has been shown to increase with road slipperiness (Norman et al., 2000), and crash severity has been found to be higher on curves (Donmez and Liu, 2015). Overall, higher motor control difficulty environments were characterized by non-dry road surface conditions or non-straight road alignments.

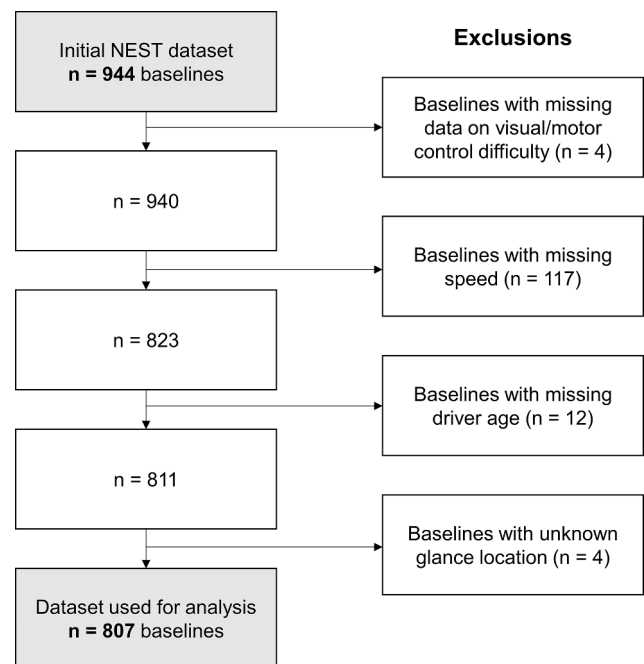
Driver Age and GPS Speed. As noted earlier, the self-regulation of secondary task engagement and visual attention allocation based on task demands may be moderated by driver age. The driving demands experienced by drivers also depend on the speed of travel, which may in turn influence secondary task engagement and glance behaviors. Thus, in addition to the two environmental demand variables, driver age and GPS speed at the beginning of each baseline were also included in our models. In the NEST dataset, driver age is categorized into 15 levels, ranging between 16 and 89. To aid statistical power, we aggregated these 15 levels into five: 16–19 ($n = 49$), 20–24 ($n = 69$), 25–34 ($n = 31$), 35–64 ($n = 26$), 65 and over ($n = 26$).

2.3. Data pre-processing

As mentioned earlier, the NEST dataset consists of 944 baselines. However, some records were excluded due to missing information as outlined in Fig. 1. The dataset used for statistical analysis consisted of 807 baselines: 613 with secondary task engagement and 194 without.

2.4. Statistical models

Statistical analyses were carried out in SAS University Edition. All models started with the following explanatory variables: visual difficulty (lower, higher), motor control difficulty (lower, higher), driver age

**Fig. 1.** Data exclusions.

(16–19, 20–24, 25–34, 35–64, or 65 and over), GPS speed at the start of the baseline, and all two-way interactions of these variables. Backward selection was used to drop non-significant interaction terms from the final models. Secondary task engagement, the dichotomous variable indicating whether or not the driver engaged in a secondary task, was analyzed in a logistic regression (or logit) model. Percentage of time metrics (i.e., for eyes-off-path and for eyes in a non-driving-related direction) were analyzed with mixed linear models. A transformation was performed on percentage of time metrics to satisfy model assumptions as follows: *logarithm* (percent time + 1). Total number of glances was analyzed through Poisson models; the logarithm of the baseline duration was introduced as an offset variable given the slight variability observed across the length of different baselines (varied around 20 s). As for number of glances longer than 1.6 s and number of glances longer than 2 s, the outcome variables had to be grouped into bins (e.g., 0, 1, and ≥ 2)

because long off-path glances and long non-driving-related glances were infrequent. Logit or ordered logit models were built for these variables. For logit, Poisson, and ordered logit models, repeated measures were accounted for through generalized estimating equations. For mixed models, participant was a random factor.

3. Results

As stated earlier, the dataset that was analyzed consisted of 807 baselines. The number of baselines with and without secondary task engagement across different environmental demand levels is presented in Table 4. Modeling results for secondary task engagement and off-path glances are presented in Table 5.

3.1. Secondary task engagement

The results of the logit model suggest that drivers adapt their secondary task engagement based on environmental demands and the speeds they are driving at. Higher visual difficulty was associated with decreased likelihood of secondary task engagement (Odds Ratio, OR = 0.65, 95% CI = 0.43, 0.97, $p = .04$). Speed and motor control difficulty had a marginally statistically significant interaction ($p < .1$). Follow-up contrasts revealed that a speed increase is associated with a decrease in the likelihood of secondary task engagement in situations with higher motor control difficulty (for an increase in speed of 10 km/h, OR = 0.88, 95% CI = 0.79, 0.98, $p = .03$) but not those with lower motor control difficulty (for an increase in speed of 10 km/h, OR = 0.99, 95% CI = 0.92, 1.07, $p = .85$).

Given sample size limitations, we were not able to statistically analyze the relationship between environmental demand and engagement in different types of secondary tasks. However, Table 6 presents the top five most commonly observed secondary tasks for each level of environmental demand. One noteworthy observation is that “texting on

cell phone” is among the top five in all but one level with high visual and high motor control difficulty.

3.2. Glance behavior

3.2.1. Off-path glances

Higher visual difficulty was associated with a decrease in the percentage of time looking off-path (about a 3% decrease as can be seen in Fig. 2). Rate of off-path glances was estimated to be 11% lower in higher visual difficulty conditions (95% CI = 1, 20, $p = .04$), see Fig. 3(a). In other words, the rate of off-path glances in the higher visual difficulty conditions was 89% of that in the lower difficulty conditions.

As shown in Fig. 4 (top), higher visual difficulty was associated with reduced rates of long off-path glances. The percentage of baselines with 2 or more off-path glances > 1.6 s was reduced from 8.5% in lower visual difficulty to 5.9% in higher visual difficulty. There was an even greater reduction in particularly long off-path glances, i.e., those longer than 2 s. The percentage of baselines with 2 or more off-path glances > 2 s was reduced from 4.7% in lower visual difficulty to 1% in higher visual difficulty. The statistical results suggest that the odds of exhibiting more off-path glances longer than 1.6 s and more off-path glances longer than 2 s were 37% (OR = 0.63, 95% CI = 0.42, 0.94, $p = .02$) and 51% lower (OR = 0.49, 95% CI = 0.29, 0.83, $p = .008$), respectively, in higher visual difficulty situations than in lower visual difficulty situations.

As shown in Fig. 3(b), drivers 35 and over (35–64 and 65-over) exhibited reduced off-path glance rates compared to younger drivers. The statistical results show that the rate of off-path glances was 25%, 23%, and 20% lower for drivers 65 and over compared to drivers 16–19 (95% CI = 9, 37, $p = .003$), 20–24 (95% CI = 8, 36, $p = .005$), and 25–34 (95% CI = 2, 34, $p = .03$), respectively. Similarly, the rate of off-path glances was 23%, 21%, and 18% lower for drivers 35–64 compared to drivers 16–19 (95% CI = 8, 35, $p = .004$), 20–24 (95% CI = 6, 34, $p = .008$), and 25–34 (95% CI = 0.5, 32, $p = .04$), respectively.

Table 4

Number of baselines with and without secondary task engagement across different environmental demand levels. Percentages in parentheses are the proportions out of 807 total baselines.

Secondary Task Engagement	Lower visual difficulty		Higher visual difficulty	
	Lower motor control difficulty	Higher motor control difficulty	Lower motor control difficulty	Higher motor control difficulty
Yes ($n = 613$)	332 (41.14%)	134 (16.6%)	76 (9.42%)	71 (8.8%)
No ($n = 194$)	93 (11.52%)	43 (5.33%)	33 (4.09%)	25 (3.1%)
Total ($n = 807$)	425 (52.66%)	177 (21.93%)	109 (13.51%)	96 (11.9%)

Table 5

Model statistics for secondary task engagement and off-path glances.

Dependent Variables	Explanatory Variables	df	F/χ^2 -value	p -value
Secondary task engagement (yes vs. no)	Visual difficulty	1	4.46	0.04
	Motor control difficulty	1	2.53	0.12
	Driver age	4	6.34	0.18
	Speed	1	3.46	0.06
	Speed * Motor control difficulty	1	3.50	0.06
% time off-path	Visual difficulty	1796	8.39	0.004
	Motor control difficulty	1796	2.04	0.15
	Driver age	4202	1.47	0.21
	Speed	1796	0.02	0.89
	Visual difficulty	1	4.33	0.04
Rate of off-path glances (per minute)	Motor control difficulty	1	0.93	0.34
	Driver age	4	16.8	0.002
	Speed	1	0.64	0.43
	Visual difficulty	1	5.24	0.02
	Motor control difficulty	1	0.08	0.78
Number of glances > 1.6 s (0, 1, ≥ 2)	Driver age	4	8.56	0.07
	Speed	1	3.65	0.06
	Visual difficulty	1	6.95	0.008
	Motor control difficulty	1	1.27	0.26
	Driver age	4	8.54	0.07
Number of glances > 2 s (0, 1, ≥ 2)	Speed	1	4.98	0.03

Table 6

The top five most commonly observed secondary tasks for different environmental demand levels. Percentage reported represents the percentage of baselines for a given demand level where a particular secondary task is observed.

Environmental Demand	NEST Secondary Task	% of Baselines
Lower visual difficulty Lower motor control difficulty <i>n</i> = 488 baselines	Passenger in adjacent seat - interaction	19.3
	Talking/singing	18.2
	Looking at an object external to the vehicle	11.7
	Texting on cell phone	9.8
	Cell phone, other	9.0
Lower visual difficulty Higher motor control difficulty <i>n</i> = 207 baselines	Passenger in adjacent seat - interaction	18.8
	Talking/singing	14.5
	Other external distraction	10.6
	Texting on cell phone	9.2
	Adjusting/monitoring radio	8.7
Higher visual difficulty Lower motor control difficulty <i>n</i> = 137 baselines	Talking/singing	25.5
	Passenger in adjacent seat - interaction	13.9
	Cell phone, other	11.7
	Adjusting/monitoring radio	9.5
	Texting on cell phone	7.3
Higher visual difficulty Higher motor control difficulty <i>n</i> = 108 baselines	Talking/singing	17.6
	Passenger in adjacent seat - interaction	14.8
	Cell phone, other	13.9
	Adjusting/monitoring radio	11.1
	Object in vehicle, other	9.3

As shown in Fig. 4 (bottom), drivers 65 and older had the lowest rate of long off-path glances, exhibiting 2 or more off-path glances > 2 s in only 1.1% of baselines. The youngest age group also exhibited a relatively low rate of long off-path glances, in particular, glances longer than 2 s (1.1% of their baselines had 2 or more glances > 2 s). Drivers between the ages of 25–34 and 35–64 had the highest and second-highest rates of long off-path glances, respectively. Statistical results suggest that driver age was marginally statistically significant for long off-path glances. Drivers 65 and over were less likely to exhibit long off-path glances (>1.6 s and > 2 s) compared to drivers 25–34 (1.6 s: OR = 0.44, 95% CI = 0.21, 0.93, $p = .03$; 2 s: OR = 0.36, 95% CI = 0.16, 0.82, $p = .02$) and 35–64 (1.6 s: OR = 0.52, 95% CI = 0.26, 1.1, $p = .07$; 2 s:

OR = 0.43, 95% CI = 0.19, 0.99, $p = .048$).

Higher speed was associated with reduced long off-path glances. An increase of 10 km/h in speed was associated with a statistically significant decrease of 8% in the odds of exhibiting more off-path glances longer than 2 s (OR = 0.92, 95% CI = 0.86, 0.99, $p = .03$), and a marginally statistically significant decrease of 6% in the odds of exhibiting more off-path glances longer than 1.6 s (OR = 0.94, 95% CI = 0.89, 1.00, $p = .056$).

3.2.2. Non-driving-related glances

As discussed earlier, although there were some ambiguities in categorizing glances as driving-related or non-driving-related, we wanted to explore non-driving-related glance behaviors based on what was possible with the NEST dataset. Overall, only age had a significant association with non-driving-related glances (percent time: $F(4, 217) = 3.88$, $p = .005$; glance rate: $\chi^2(4) = 29.08$, $p < .0001$; and rate of glances > 1.6 s: $\chi^2(4) = 9.93$, $p = .04$).

Older drivers (aged 65 and above) spent proportionally the least time looking at non-driving-related directions. Follow up contrasts revealed that compared to drivers aged 65 and older, the percent of time spent looking at non-driving-related directions was lower for drivers aged 16–19, 20–24, and 25–34. Overall, drivers 35 and over exhibited lower rates of non-driving-related glances compared to younger drivers. In particular, the rate of non-driving-related glances was 64%, 62%, and 68% lower for drivers 65 and over compared to drivers 16–19 (95% CI = 43, 78, $p < .0001$), 20–24 (95% CI = 41, 76, $p < .0001$), and 25–34 (95% CI = 44, 82, $p = .0001$), respectively. Similarly, the rate of non-driving related glances was 47%, 45%, and 53% lower for drivers 35–64 compared to drivers 16–19 (95% CI = 17, 66, $p = .006$), 20–24 (95% CI = 14, 64, $p = .009$), and 25–34 (95% CI = 17, 74, $p = .009$), respectively. Drivers 65 and over were less likely to exhibit non-driving-related glances longer than 1.6 s compared to drivers 16–19 (OR = 0.11, 95% CI = 0.02, 0.90, $p = .04$), 20–24 (OR = 0.11, 95% CI = 0.02, 0.85, $p = .03$), 25–34 (OR = 0.05, 95% CI = 0.01, 0.43, $p = .006$), and 35–64 (OR = 0.14, 95% CI = 0.02, 1.04, $p = .054$). Overall, these findings on age-related differences were in line with our off-path glance results.

4. Discussion

4.1. Summary of key findings

Our analysis of naturalistic driving data from the NEST dataset suggests that drivers adapt their secondary task engagement and glance

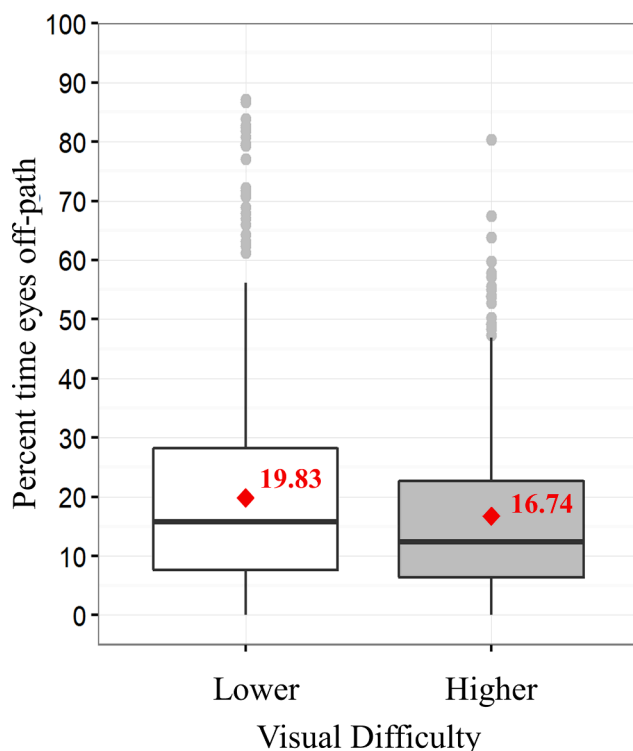


Fig. 2. Boxplots showing quartile information and means via red diamonds for percent of time eyes were off-path.

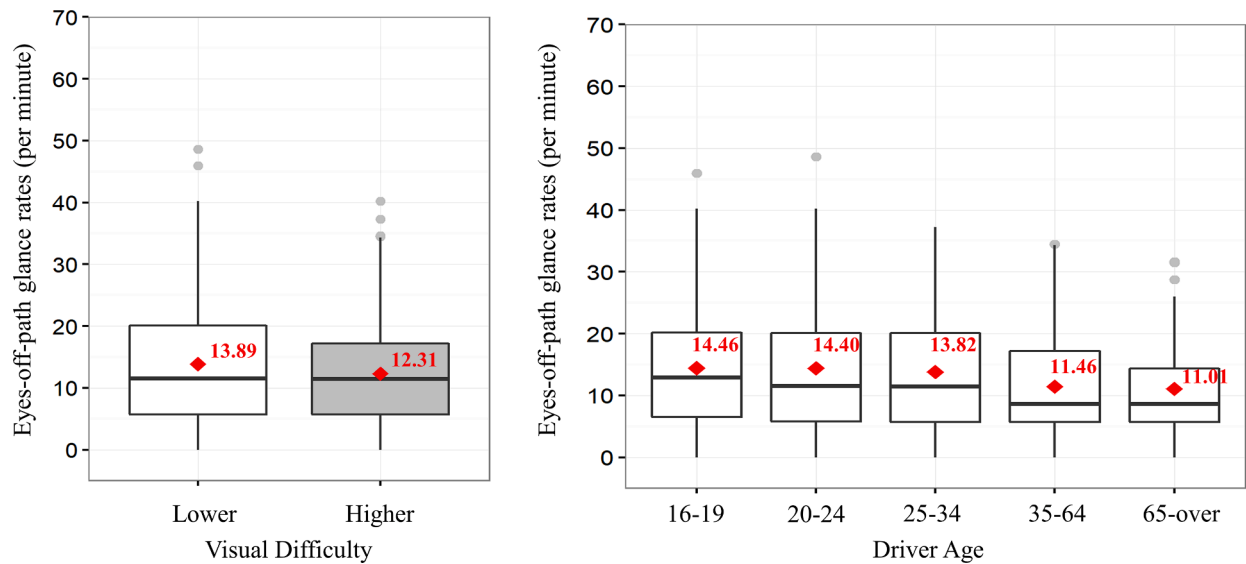


Fig. 3. Boxplots showing quartile information and means via red diamonds for rate of off-path glances per minute across (a) visual difficulty, and (b) driver age. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

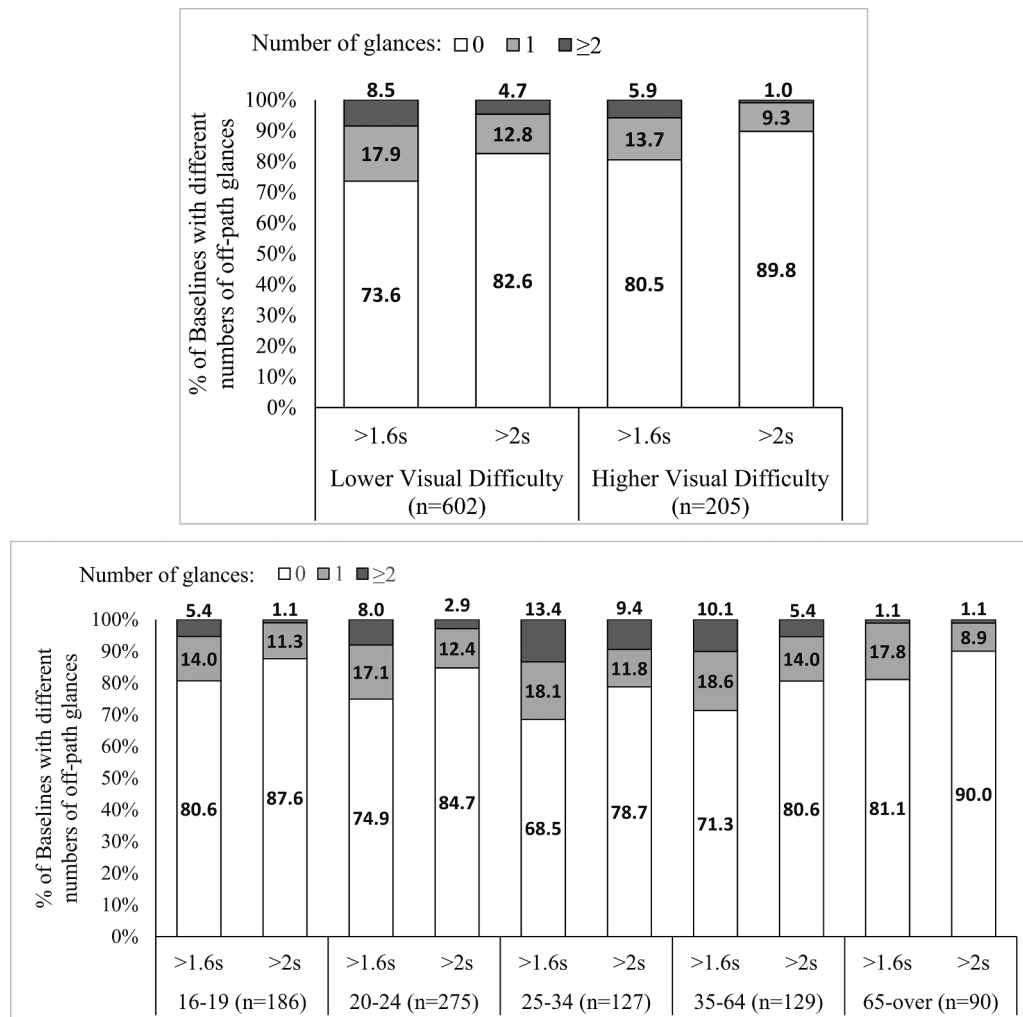


Fig. 4. Distribution of long off-path glances across (top) visual difficulty and (bottom) age groups.

behaviors based on driving demands. In particular, the likelihood of secondary task engagement decreased in environmental conditions of higher visual difficulty. Higher visual difficulty was also associated with

a lower percentage of time looking off-path (about 3% less), lower frequency of off-path glances (about 1.5 fewer glances per minute), and lower likelihood of exhibiting long off-path glances (>2s and > 1.6 s).

Long off-path glances (>2s) were infrequent in general, which is promising given their link to increased crash risk (Klauer et al., 2006; Simons-Morton et al., 2014), but they were even less frequent in situations with higher visual difficulty. For lower visual difficulty conditions, 17.5% of baselines had at least one glance longer than 2 s compared to 10.3% for higher visual difficulty conditions. Further, off-path glances longer than 2 s were reduced to a greater extent in more visually difficult situations than those longer than 1.6 s. Our investigation of the most frequent secondary tasks can also provide further indication that drivers reduce potentially risky behaviors based on driving task demands, given that texting was among the top 5 most frequent tasks in all conditions except for the high visual and high motor control difficulty category. Texting is known to be one of the most dangerous secondary tasks (Dingus et al., 2016).

An increase in speed was associated with a reduction in long off-path glances (>1.6 s and > 2 s): off-path glances longer than 2 s were reduced to a slightly greater extent. There was also a marginally statistically significant interaction between speed and motor control difficulty for secondary task engagement: speed was found to be associated with a decrease in the likelihood of secondary task engagement in higher motor control difficulty situations. These findings suggest that increase in speed is associated with reductions in potentially risky behaviors.

In general, we found associations between age and rate of off-path glances, as well as age and metrics for non-driving-related glances. Drivers 35 and older had reduced rates of off-path glances compared to those who were younger (about 2–3 fewer off-path glances per minute). It may be that middle-aged and older drivers are less inclined to engage in visual secondary tasks, many of which are technology-related (e.g., texting) as categorized in NEST (see Table 2). They may also have better self-regulation as they may assess risk more accurately. However, because we examined secondary task engagement in general (not only visual secondary tasks), we did not observe the same relationship between age and secondary task engagement. Finally, we note that age is known to correlate with driving skills (Carr et al., 1992) and information processing abilities (Maltz and Shinar, 1999; Panek et al., 1977), and is likely a proxy for skills and cognitive abilities in our models. Further research is needed to provide deeper insights into the moderating effects of skills and abilities when investigating the relationship between driving demands and secondary task engagement.

4.2. Contributions and limitations

Our findings on the likelihood of secondary task engagement in different environmental conditions contribute to the literature on driver distraction, especially on understanding the role of self-regulation in distraction engagement. Earlier naturalistic studies have identified similar adaptive behaviors related to cell-phone tasks (Funkhouser and Sayer, 2012; Tivesten and Dozza, 2015). The current study is a more general investigation of drivers' regulation of their secondary task engagement and thus provide insights on activities that potentially distract drivers without taking their eyes off the road. Our analysis of off-path glances provided a more specific investigation of whether and how drivers self-regulate particularly risky behaviors—e.g., long off-path glances—based on the demands of the environment. Findings on glance behaviors are in line with previous on-road and naturalistic studies that investigated driver glance behaviors (Räsänen and Summala, 1998; Shinar, 2008; Tijerina et al., 2004; Tivesten and Dozza, 2014). For example, we found that drivers reduced their longer off-path glances (>2s), which can be particularly risky (Klauer et al., 2006; Simons-Morton et al., 2014), to a larger extent than shorter ones (>1.6 s). Similarly, Tivesten and Dozza (2014) found that higher driving demands resulted in shorter off-road glances.

It should be noted that we focused on off-path glances, rather than distraction-related glances, given that the glance coding in NEST was

limited to the direction of the glance, not the specific context. In other words, we were not able to identify situations where the driver looked off-path but was glancing at elements in the environment that were driving-related. Acknowledging this limitation, we reported an additional analysis of glance behaviors using glances that are classified as potentially relevant to driving.

Our categorization of environmental demands was exploratory in nature and may not be entirely accurate. Some of the variables used to label visual difficulty (e.g., level of service) may be confounded with motor control difficulty. For example, traffic density is expected to increase visual complexity, but it may also increase motor control difficulty due to the need for more braking responses in heavier traffic (Brookhuis et al., 1991; De Waard et al., 2008). Future work may attempt different combinations of variables to represent levels of environmental demands, or include more variables in the analysis. The current analysis is however limited by sample size available from the NEST dataset.

There is a need to better characterize environmental demands to provide a better context for existing distraction detection algorithms (e.g., Vanysek et al., 2005) and driver support systems (e.g., Trivedi and Cheng, 2007). Given that the environmental variables provided in NEST are also available in the larger SHRP2 data, future analysis of the SHRP2 data can also utilize our approach to examine the impact of environmental demands on driving in general. One aspect of self-regulation that we could not consider was the demand imposed by the secondary task; this was due to a lack of more specific information on the secondary tasks in the NEST data. Although we have captured drivers' self-regulation of their general secondary task engagement based on driving demands, it would have been interesting to investigate whether drivers adapted their engagement behaviors differently for different secondary tasks.

Due to missing information, a considerable number of NEST baselines were dropped from this analysis (missing speed information, in particular, resulted in the loss of 117 baselines from the dataset). Our findings should be interpreted with the missing baselines in mind, as they may result in a loss of power in our analysis, and a potential for systematic bias if the missing data is not random. More importantly, as stated earlier, NEST data only includes drivers who were involved in at least one distraction-related crash or near crash. Thus, they are not representative of the entire driving population, but one that is particularly at-risk for distraction-affected safety critical events.

Our findings suggest that drivers may adapt their secondary task engagement and visual attention allocation based on environmental demands in a potentially risk-reducing manner. However, further research is needed to translate the effects we observed in task engagement and glance behaviors to actual crash risk savings. Further, there is still a need for interventions that can mitigate the effects of distractions (e.g., Donmez et al., 2007) given that some secondary task engagement, as well as inopportune short off-road glances, have been shown to increase crash risks (Dingus et al., 2016; Victor et al., 2015). Understanding how individuals modulate their distraction engagement and visual attention allocation based on driving demands can help inform such interventions.

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